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Demagnetization of Igneous Rocks by Alternating Magnetic Fields†

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Abstract

The presence of secondary components of magnetization in basalts often limits the usefulness of their directions of natural remanent magnetization for palaeomagnetic work. These secondary components may be removed by alternating magnetic fields. The procedures for doing this are described and tests based on the internal consistency of the results are devised to judge their reliability. These are applied successfully to a series of results from specimens collected from the Tertiary basalts of New South Wales, Australia, and a determination of the geomagnetic field in S.E. Australia during the Lower Tertiary is obtained. A study of the behaviour of these basalts in high alternating fields suggests that most of the iron mineral grains have a multi-domain character, and contain a hard and soft component in the ratio of about 1 : 10.

§ 1. Introduction

In order to obtain the direction of the ancient geomagnetic field from a study of rock magnetism it is necessary to measure the direction of the primary component of magnetization acquired at the time the rock is formed. This primary magnetization is often seriously affected by secondary components, which are added later, and which render the direction of the *natural remanent magnetization* (NRM) of little use for precise palaeomagnetic work.

Lavas, on cooling through their Curie point, acquire a primary *thermo-remanent magnetization* (TRM) in the direction of the earth's field. If magnetic particles with relaxation times less than the geological age of the rock are present, a secondary component is acquired different in direction from the primary one if the direction of the geomagnetic field has changed since the latter was acquired. This is viscous magnetization and such components have the form of an *isothermal remanent magnetization* (IRM). Rock specimens containing grains with relaxation times in the earth's field of the order of 10^2 or 10^3 years will contain components of magnetization along the present dipole field. Relaxation times of 1 to 100 years will produce magnetizations close to the present field, whereas shorter relaxation times produce further viscous components dependent on the conditions of storage. This latter type of secondary components

† Communicated by the Authors.

has been termed temporary components by Creer (1957). Secondary components are the most substantial source of error in palaeomagnetic studies and may be removed by alternating magnetic fields. The problem is to find the alternating field conditions necessary for their removal and yet ensure that a measurable component of primary origin remains. Because of the variation in magnetic minerals the field strength will not necessarily be the same for all lava types even within the same formation.

In general the removal of unwanted secondary components from specimens collected from a lava sequence will decrease the scatter in the directions of magnetization. We suggest that certain consistency conditions should be fulfilled before the secondary components may be judged to have been removed. Compared with the directions of NRM ($\mathbf{M}_n$) the directions after removal of the secondary components should have these characteristics.

(1) The scatter between specimens taken from the same lava flow (within-site dispersion) should either remain constant or should decrease.

(2) The agreement between the mean directions of different lava flows (between-site dispersion) should be much improved.

(3) Assuming the average dipole character of the earth's field the mean direction of magnetization for all flows should agree with similar results from rock formations of comparable age in the same region after correction for differences in geographical location.

These conditions apply to the great majority of lava sequences which were laid down in a period of time during which the geomagnetic field had an approximately constant direction (irrespective of sign) different from the present field. The first will in most cases be necessary, but it is possible to envisage situations in which the within-site scatter increases slightly after removal of the secondary component. The second is a necessary and the most important condition. The third is also necessary given the stated assumption. As a test of these conditions basalt specimens from several lava flows have been treated in alternating magnetic fields (§§ 2, 3, 4 and 5).

Further experiments on the same specimens were carried out in alternating magnetic fields to study the behaviour after removal of most of the primary magnetization and it was not found possible to reduce the moment beyond a certain limit, called the *minimum intensity*. This property, which may be of some interest in the theory of NRM of basalts, is discussed in § 7. In § 8 an attempt is made to relate the primary and secondary components to the ferromagnetic grains present.

2. Demagnetization by Alternating Magnetic Fields

The procedure is to subject a rock specimen (in the absence of a steady field) to an alternating field which is reduced smoothly. The demagnetizing instrument used is similar to that recently described by Creer (1959). Specimens are placed in a holder which spins inside a set of coils which

annul the earth's field. The alternating field is provided by a solenoid mounted on a trolley running on an aluminium tramway. The field at the specimen is reduced by withdrawing the trolley to a distance of 3 m at which point the field is reduced by a factor of 3×10^5.

Two pairs of square coils (Parry 1957) side length 180·2 and 165·9 cm and separations 98·2 and 90·4 cm and each with 100 turns annul the horizontal and vertical components of the earth's field respectively. Chokes are included in both circuits to reduce the effect of any mutual inductance with the solenoid, which in any case should be zero since the solenoid axis and tramway are set perpendicular to the magnetic meridian and to the axes of the Parry coils. The demagnetizing solenoid (i.d. 14·7 cm, o.d. 20·8 cm, length 12·5 cm) contains 1920 turns in 20 layers of 18 s.w.g. enamelled copper wire and is similar to that described by Parry (1957). The resistance is 15·6 Ω and inductance 0·5 H. The coil has been calibrated by a.c. and d.c. methods and both methods agree to 1% and give a calibration of 105 oersteds/amp. Power is provided by public supply mains (240 v; 50 c/s) through a bank of condensers (20 μF), which, with the coil, form a circuit tuned to 50 cycles. This reduces undesirable harmonics and as an extra precaution a series-resonant circuit tuned to 100 cycles has been connected in parallel. The required voltage is obtained from a variable transformer. The rotations about the vertical and horizontal axes are in the ratio 15 : 16. Gears provide primary rotation rates about the vertical axis of 140, 350, 525 and 700 rev/min, which do not resonate with any simple fraction of the mains frequency. It was found experimentally that the results are unaffected by changes within this range and the lowest rate has been used in these experiments.

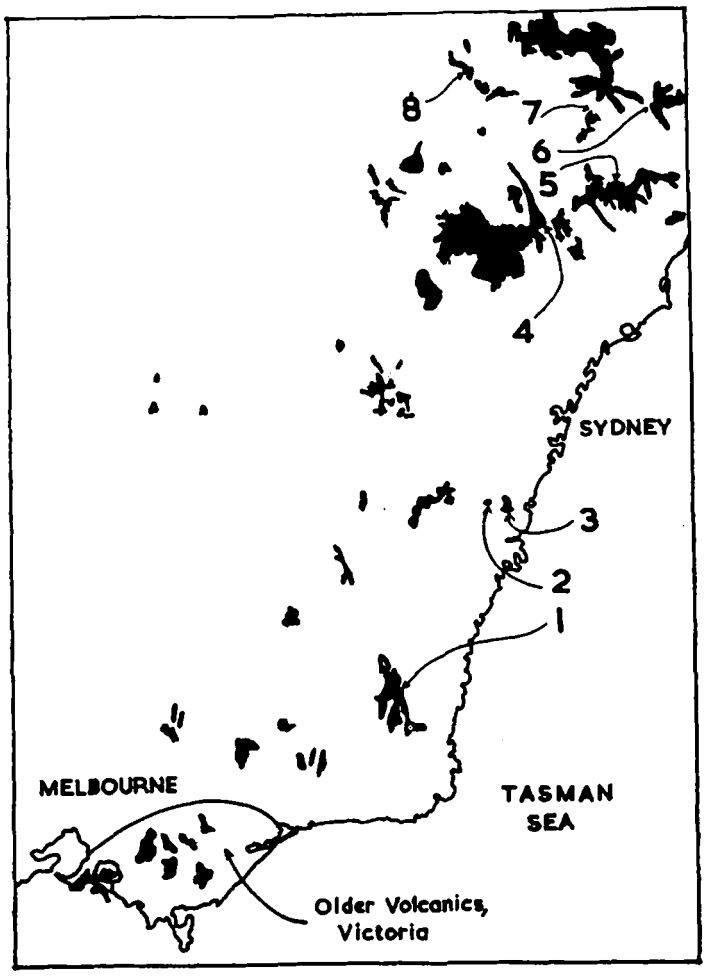
The direction and intensity of magnetization of each specimen is first measured using an astatic magnetometer. Directions are determined to 3° and intensity to 5%. The specimen is treated in a small alternating field and then remeasured. Treatment in a higher field is then given and further measurements taken, and so on. The peak values ($H_p \sim$) of the alternating fields used are always quoted. The field strengths generally used are 75, 150, 300, 450, 600 and 750 oersteds, although in many cases intermediate steps are included. Provided the field used is less than that required to reduce the specimen to its minimum intensity (a situation which is explained later) repeat treatments give the same results.

§ 3. Rock Specimens Studied

The rock specimens chosen for study were collected from eight sites (fig. 1) in the Tertiary Basalts of New South Wales (see general account of geology in David (1950)). The specimens were chosen because they showed different degrees of magnetic stability. The NRM consists of a primary component together with secondary components of varying magnitude directed approximately along the present geomagnetic field or along the field during storage in the laboratory.

Natural exposures in these basalts are always extensively weathered and the collections have been made in fresh exposures in quarries and

Fig. 1



Locality sketch-map. The basalt outcrop is in solid black. The collecting localities are numbered 1 to 8 as in the text and tables. The Older Volcanics of Victoria which has been studied previously are also marked.

1. Road cutting 2 km N of Nimmitabel on Monaro Highway.

2. Road cutting on Hume Highway, 5 km NE of Berrima (Site B, Green and Irving 1958), (Map reference : Mittagong 370437).

3. Three sites, one in Robertson village (Kiama 627294), one on Robertson-Moss Vale Road, 3 km W of Robertson (Kiama 560284) and the third on Belmore Falls Road (Kiama 594275).

4. Road cutting, Nundle–Timor Road, 18 km SW of Nundle.

5. Road cutting on Oxley Highway, 53 km E of Walcha.

6. Road cutting on the Guyra Road, 5 km S of Ebor.

7. Two quarries, one on site of University of New England and the second 2½ km NW of Armidale.

8. Road cutting 50 m W of bridge over Myall Creek on Delungra–Bingara Road, 24 km E of Bingara.

Table 1. Mean site directions and poles. The collecting sites are numbered in the text and figures. S is the number of samples and N is the number of specimens cut from these. Mean site directions of NRM and the mean directions after removal of the secondary and temporary components are given. The lavas are flat-lying and no correction for geological tilt is necessary. D is the declination east of true north, I is the inclination positive downwards, R is the resultant and k is Fisher's (1953) precision parameter. The mean pole is 63 S, 137 E ($N = 8$, $R = 7{\cdot}21$, $\alpha(P0{\cdot}05) = 20°$)

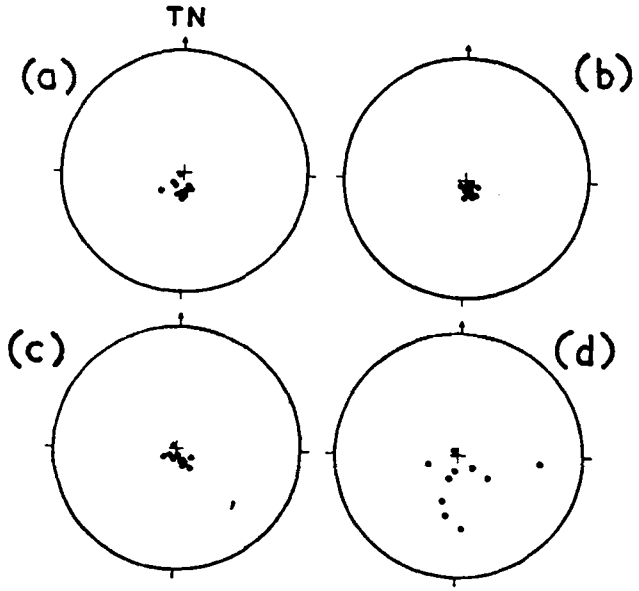
Collecting sites	Lat. S. Long. E.	S	N	Mean directions of NRM				Mean directions after treatment in alternating fields				Pole position (primary magnetization)	
				D	I	R	k	D	I	R	k	Lat. S	Long.
1. Nimmitabel	36° 30′ 149° 16′	2	4	167	+59	3·50	6	166	+88	3·63	8	40	150E
2. Berrima	34° 28′ 150° 23′	7	11	187	+76	10·78	44	164	+79	10·92	177	54	160E
3. Robertson	34° 36′ 150° 34′	6	11	Random		3·43	—	240	+75	8·43	4	44	117E
4. Liverpool Ranges	31° 38′ 151° 41′	2	4	57	−12	3·94	54	140	+47	3·92	41	55	126W
5. Yarrowitch	31° 15′ 151° 58′	7	16	274	+46	13·32	6	219	+60	15·35	23	57	91E
6. Ebor	30° 25′ 152° 20′	3	5	190	+60	4·41	7	191	+62	4·53	8	74	122E
7. Armidale	30° 31′ 151° 39′	6	10	166	− 3	7·78	4	199	+45	9·46	17	73	55E
8. Myall Creek	29° 47′ 150° 44′	3	5	92	+84	4·94	65	209	+82	4·88	33	44	140E

road-cuttings spread over a meridional distance of 650 km. The basalts are flat-lying and no geological correction to the directions of magnetization is necessary. In six cases, each site is a quarry or road-cutting in which the samples cover 5–10 m in a single flow. At Armidale (Site 7) samples were obtained from two quarries about 1 km apart. At 3, three road-cuttings 2–3 km apart west of the village of Robertson were sampled. At both these sites more than one flow may be represented, although if this is so they cannot be much separated in time. Several separately orientated samples have been taken at each site and from each sample one or more cylindrical specimens (3·5 cm diameter, 3·5 cm height) have been cut. The details are given in table 1. Altogether there are 37 samples and 66 specimens.

§ 4. Results

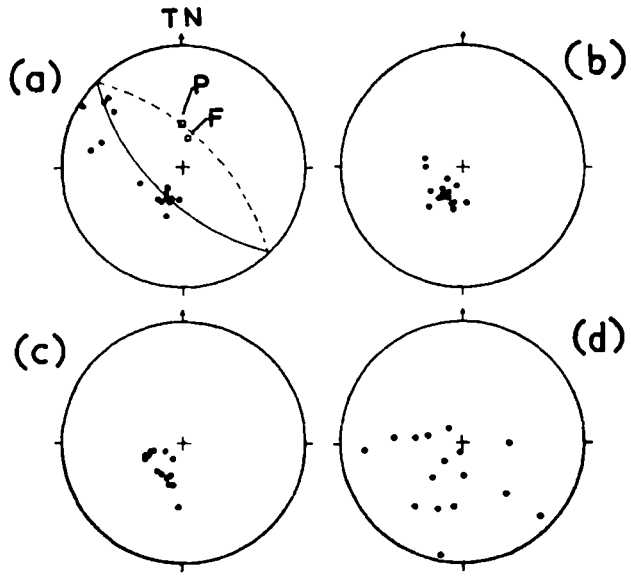
The demagnetization procedure has been carried out on all specimens, about 500 determinations of the direction and intensity of magnetization being necessary. The effect on the directions of magnetization in specimens from the same site is shown for two sites in figs. 2 and 3. The variation in the within-site dispersion may be illustrated by plotting k/k_{81g} against the

Fig. 2



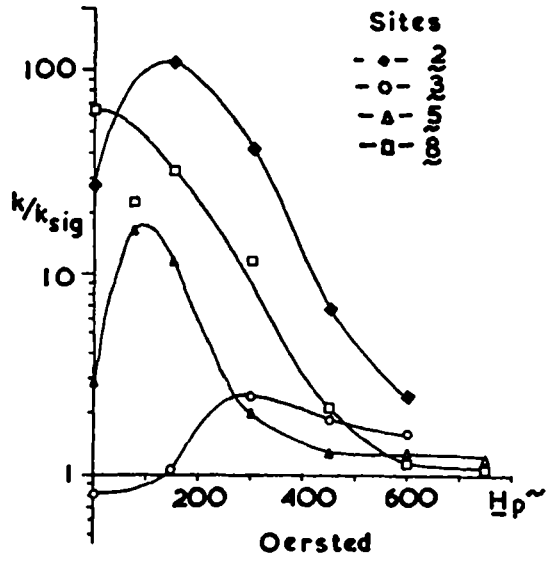
Effect of alternating fields on the directions of magnetization at site 2. (*a*) NRM, and after treatment in (*b*) 150, (*c*) 300, and (*d*) 450 oersteds. The projections used in this and figs. 3, 7 and 8 are polar stereographic. North-seeking directions are plotted as dots on the lower hemisphere or circles on the upper hemisphere. The inclination scale is given in 7 (*a*).

Fig. 3



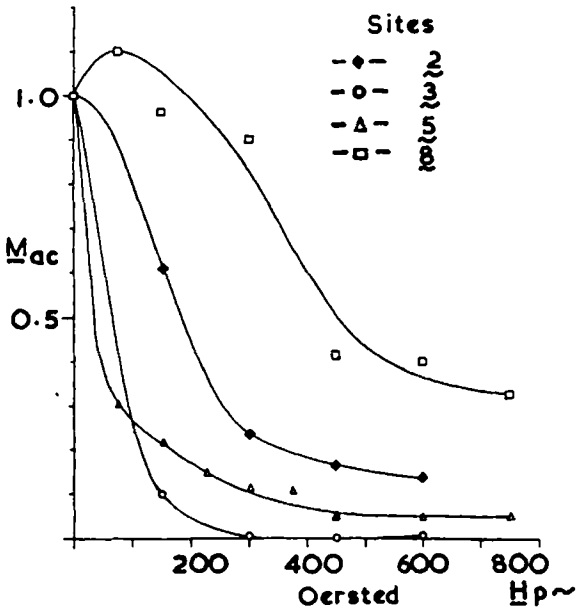
Effect of alternating fields on the directions of magnetization at site **5**. (*a*) NRM, and after treatment in (*b*) 75, (*c*) 150 and (*d*) 750 oersteds. In (*a*) **P** is the dipole field direction and **F** the present field both on the upper hemisphere. The great circle (continuous on the lower hemisphere and dotted on the upper) through **P** and the mean direction after treatment in 75 oersteds is given in (*a*) and shows the ' stringing ' of $\mathbf{M_n}$.

Fig. 4



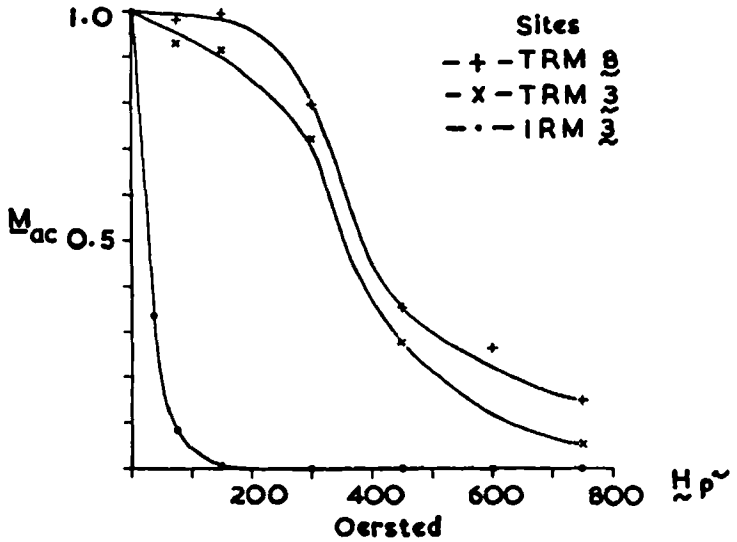
Effect of alternating magnetic fields on the within-site dispersion. This is illustrated by plotting the parameter k/k_{sig} (see text), which is proportional to the inverse of the dispersion, against alternating field. The horizontal axis is drawn at $k/k_{sig} = 1$ and values below this indicate a random within-site dispersion.

Fig. 5



Demagnetization curves for specimens from sites **2, 3, 5** and **8.** The value M_{ac} is plotted against alternating field $H_p\sim$ and M_{ac} is the intensity after treatment, normalized to $M_{ac}=1$ at $H_p\sim=0$.

Fig. 6



Demagnetization of IRM and TRM. The value of M_{ac} is plotted as in fig. 5. The initial intensities of TRM in 0·6 oersted ($M_{t(0\text{-}6)}$) are $0{\cdot}54\times10^{-3}$ e.m.u./cm³ for a specimen from **8** and $22{\cdot}5\times10^{-3}$ for a specimen from **3.** The IRM ($M_{i(300)}$) acquired by another specimen from **2** in a field of 300 oersteds is 108×10^{-3}.

strength of applied alternating field. This has been done for all sites and representative curves for four sites are given in fig. 4. The parameter k is an estimate of κ the dispersion parameter occurring in the direction distribution $\exp(\kappa \cos\theta)$ (Fisher 1953), and $k_{sig} = (N-1)/(N-R_0)$ where R_0 is the value, tabulated by Watson (1956), of the resultant of N random vectors which will be exceeded in only 5% of cases. When $k/k_{sig} \leqslant 1$ the distribution may be considered random.

The associated variations in intensity are illustrated by results from specimens from the same four sites (fig. 5). For comparison the demagnetization of IRM and TRM are given in fig. 6. The specimens were first demagnetized in 900 oersteds alternating field and then magnetized in 300 oersteds steady field to give $M_{i(300)}$, or heated in nitrogen to 625°C and cooled in the earth's field (0·6 oersted) to induce $M_{t(0\cdot6)}$.

Table 2. Some magnetic properties. M_n is the intensity of the natural remanent magnetization M_{min} is the minimum intensity (see § 7), $M_{i(sat)}$ is the saturation isothermal remanence, H_{sat} is the field required to saturate and H_{cr} is the coercivity of remanence of $M_{i(sat)}$ obtained by the methods described by Creer *et al.* (1959). Intensities are given in units of 10^{-3} e.m.u./cm^3 and field strengths in oersteds

Site number	M_n	M_{min}	$M_{i(sat)}$	H_{sat}	H_{cr}
1	10·48	0·06	101·8	900	200
2	5·95	0·82	188·0	1700	225
3	32·25	1·34	224·5	1000	205
4	13·74	0·26	354·0	1500	190
5	2·52	0·24	354·0	900	180
6	1·25	0·11	405·0	1200	180
7	7·13	0·86	236·5	900	160
8	0·014	0·005	0·32	4000	600

Certain other magnetic properties are listed in table 2: amongst these it may be noted that the 'back field' (the coercivity of remanence, H_{cr}) required to remove the saturation isothermal remanent magnetization ($M_{i(sat)}$) is usually about 200 oersteds.

These results are now discussed dealing first with the effects in low alternating fields.

§ 5. Effects in Low Alternating Fields ($H_{\mu} \sim < H_{cr}$)

The results within each site which are relevant to the first condition in § 1 are considered first. At site **8** the directions of NRM ($\mathbf{M}_n$) have large positive inclinations. The directions are well grouped and this is not improved by treatment in alternating fields. In fields up to 400 oersteds the demagnetization curve of M_n is convex upwards very similar to $M_{i(0\cdot6)}$ induced in the laboratory. The small initial rise is presumably due to the removal of a soft component, along the earth's field and which is approximately anti-parallel to the hard component. The coercivity of remanence (H_{cr}) is high (table 2). It may be concluded that the NRM is predominantly stable and of TRM type.

At site **3** the directions $\mathbf{M}_n$ are random (see fig. 5 (*b*), Boesen *et al.* 1961), but these become grouped together with steep positive inclinations after treatment in alternating magnetic fields of 300 oersteds (table 1). The demagnetization curve is convex downwards similar to that for IRM induced in the laboratory although the initial decrease is not so rapid. The directions in specimens after storage in the laboratory for three months can change by as much as 180°. The value of H_{cr} is low (table 2). We conclude that the NRM of this basalt consists of a highly unstable magnetization and a stable component of much smaller magnitude with steep positive inclination.

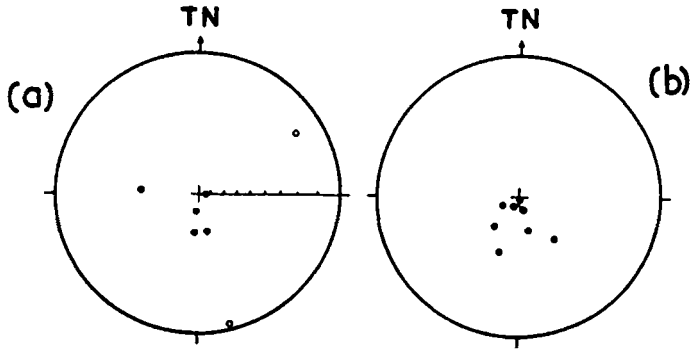
At site **2** the directions $\mathbf{M}_n$ are well grouped with steep positive inclinations (fig. 2 (*a*)). At **5** there is a group of specimens with similar directions but also there are others with low inclinations and north-westerly declinations strung out close to a great circle through the dipole field (fig. 3 (*a*)). At both sites after treatment in small alternating fields the disperion decreases and the directions all become southerly with large positive inclinations. The minimum dispersion occurs after treatment in 100–200 oersteds. The demagnetization curves lie mainly between those for **8** and **3** and between $M_{i(0\cdot6)}$ and $M_{i(300)}$ induced in the laboratory. The values of H_{cr} are low. The conclusion is that the NRM consists of a comparatively stable component of TRM-type and an unstable IRM-type component of similar or lesser magnitude. The stable component is reversed and is appreciably affected by low demagnetization fields since the demagnetization curves do not show the initial rise as at **8**. The behaviour of specimens at the other sites (**1**, **4** and **7**) is similar to that at **2** or **5**.

Turning now to the comparison of results between-sites (condition (2) in § 1) it is immediately clear that the effect of treatment in low fields (300 oersteds for site **2** and 150 oersteds for the remaining sites) is to bring the widely scattered mean directions of NRM at each site into close agreement (table 1 and fig. 7 (*a*) and (*b*)). The divergences remaining between the site mean directions are due mainly to experimental error and any residue may be satisfactorily explained as due to the secular variation in the geomagnetic field during the time spanned by the lavas.

The third condition may now be tested by comparing the directions with those previously obtained from Victoria (see fig. 1). The directions

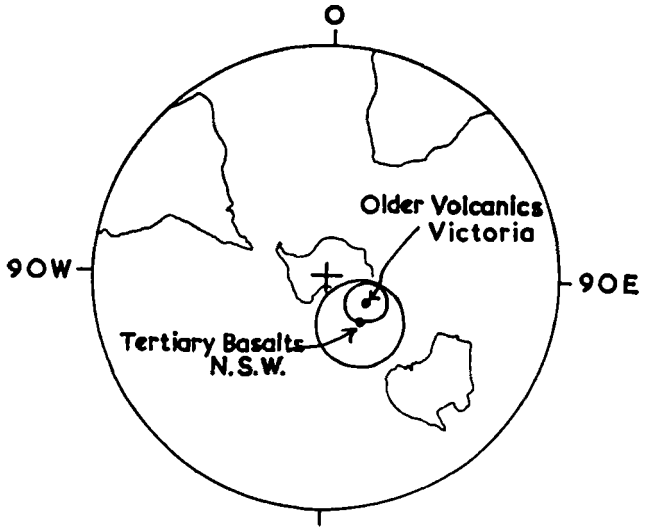
irrespective of sign are very similar to those observed in the Older Volcanics of Victoria (compare fig. 7 (*b*) with fig. 6 of Irving and Green (1957)). The poles for each site are given in table 1 and the mean of these (63 S, 137 E)

Fig. 7



Site mean directions. (*a*) NRM, (*b*) after treatment in low alternating fields. The initial within-site directions at **3** are random and there is no significant mean, but after treatment in 300 oersteds alternating field the directions become grouped together and the mean of these is included in (*b*).

Fig. 8



Pole position for the N.S.W. basalts. The pole for the Older Volcanics of Victoria is added for comparison. The ovals around the poles mark out the zones within which the pole lay at a probability of 95%. The south geographic pole is marked with a cross.

agrees well with the pole for the Older Volcanics of Victoria (67 S, 123 E) (see fig. 8). The spread of time of the New South Wales basalts is probably considerable, but does not necessarily overlap that of the Older Basalts of Victoria although both are Lower Tertiary. The Lower Tertiary sequence of Victoria has in some places been gently folded but the lavas have been sampled only at those places where the dip of adjacent sediments is 5° or less (Irving and Green 1957). Thus here is an example of agreement between pole positions obtained from rocks from the same land unit which have suffered no appreciable tectonic movement and which are of similar geological age.

It is evident that after treatment in low alternating fields the three conditions listed in § 1 are satisfied and that the effects of the secondary magnetizations imposed by the present earth's field have been removed. Since the stable magnetization, for instance, at site **8** where the basalt has not been appreciably affected by a secondary magnetization, is of TRM type, and in the absence of any other known geophysical force adequate to produce consistent directions of magnetization over a meridional distance of a 1000 km we may assume that the directions of magnetization remaining after this treatment are the directions of the primary TRM and thus of the geomagnetic field at the time the lavas cooled.

One further point which for the sake of completeness needs to be considered here is the effect of burial stress, which could conceivably have had a systematic effect on the inclination. Except for laterite and soil cover, no deposits overlie the basalts. The thickness of the lavas is not usually more than 100 m. The maximum is probably to be found in the Liverpool Ranges (fig. 1) where 500 m of lavas occur which may be taken to be not more than a factor of two less than the maximum burial depth. To test this point the magnetization of pairs of discs cut from two samples at each of sites **5** and **7** was further examined in the laboratory. In one of each pair, IRM in a field of 100 oersteds, $M_{I(100)}$ and TRM in the earth's field, $M_{t(0{\cdot}6)}$ was induced whilst the specimens were subjected to an axial load of 500 kg/cm^2, and in the other, the same magnetizations were produced without load. The apparatus and method have been described by Stott and Stacey (1960). There was no measurable deflection of the magnetization due to the application of stress. The directions of magnetization in each pair agreed to within the measurement error. The applied stress corresponds to a depth of burial of about 1500 m and it is unlikely therefore that any systematic effect due to stress during burial is present. No self-reversal of TRM was found, with or without stress.

§ 6. Procedure for Removal of Secondary Magnetization

We suggest that the following procedure may be applied to volcanic rocks.

1. At a particular site measure H_{cr} for one specimen; there is not usually much variation between specimens from the same site.

2. Measure the directions of NRM of 5 (say) specimens preferably from different samples and the directions after treating in alternating fields ($H_p \sim$) of $\frac{1}{4}H_{cr}$, $\frac{1}{2}H_{cr}$, H_{cr} and $2H_{cr}$. Find the field at which the points are least scattered, either by plotting k against $H_p \sim$ or by plotting on a stereogram.

3. Treat all specimens in this alternating field. In those cases where the primary and secondary components are of the same magnitude (that is, when the directions $\mathbf{M}_n$ are distributed along a great circle between the present field and the directions of the primary magnetization) the strength of alternating field required is usually about $\frac{2}{3}H_{cr}$. When the primary component is small compared with the secondary ones alternating fields as high as $1\frac{1}{2}H_{cr}$ may be necessary.

§ 7. Treatment in High Alternating Fields ($H_p \sim \gg H_{cr}$)

After treatment in alternating fields of 600 oersteds, which in most cases is several times greater than H_{cr}, the directions at each site approach a random distribution. The intensity does not diminish regularly with increasing alternating field, but levels off to an almost constant average value at about that field at which the dispersion begins to increase. There are however, fluctuations about this average. This value which is called the *minimum intensity* (M_{min}), is now discussed.

The effects may be due to instrumental imperfections or to inherent properties of the rocks. As and Zijderveld (1958) found that "during the experiments of demagnetization by powerful alternating fields a disturbing magnetization appears to be induced in the samples". Creer (1959), however, finds that spinning "prevents the specimens acquiring an IRM due to DC components", but even so he still finds "stray magnetizations acquired during 'cleaning'". We suggest that while the effects due to instrumental imperfections can be important if precautions are not taken, there will always be some residual, fluctuating magnetization, even when the demagnetization takes place under perfect conditions. The reason for this is that spontaneous magnetization of domains is lost only at temperatures above the Curie point, so that the effect of alternating magnetic fields is one of rearrangement of domains, and because of statistical fluctuations the resultant moment of a specimen is never likely to be zero. The observed minimum intensity, and the scattered, almost random directions in alternating fields whose strength is much greater than the coercivity of remanence may therefore be a result of such fluctuations and will depend on the number of domains in a specimen. An estimate of this number may be made as follows.

Assume firstly that there are n_1 ferromagnetic grains per cm^3 of equal size and moment M_p whose directions are random in an isotropic demagnetized rock. Rayleigh (1919) has given the probability distribution of the magnitude of the resultant r_0 of n randomly oriented unit vectors for large values of n. The value of r_0 is given by $r_0 = Cn^{1/2}$

where C is a constant having the value 1·61 for $P(r > r_0) = 0{\cdot}05$ and 0·89 for $P(r > r_0) = 0{\cdot}50$. The latter will give an average value for the mean random magnetization so that the *random moment* (M_r) per cm³ is given by $M_r = M_p \,.\, 0{\cdot}89 \,.\, n_1^{1/2}$. Assume further that in each ferromagnetic grain there are n_2 domains of equal moment m so that the total number of domains per cc is $n_1 \,.\, n_2$. We may say that domains have randomly arranged directions in a single demagnetized grain and the average moment of each grain will thus be $M_p = m \,.\, 0{\cdot}89 \,.\, n_2^{1/2}$. However it is more likely that the domains behave as an assembly of anti-parallel vectors having the values $+1$ and -1 with equal probability. Rayleigh (1880) has given the probability for the length of the resultant in this situation also and we obtain $r_0 = An^{1/2}$ where $A = 0{\cdot}83$ for $P(r > r_0) = 0{\cdot}50$. The average moment per demagnetized grain in this case will be $M_p = m \,.\, 0{\cdot}83 \,.\, n_2^{1/2}$. The correct distribution probably lies somewhere between the random and anti-parallel situations but the difference for these purposes is negligible. Taking an average for the constant as 0·86 the random moment per cc becomes:

$$M_r = m \,.\, 0{\cdot}76 \,.\, (n_1 \,.\, n_2)^{1/2}. \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (1)$$

Now the saturation moment $M_s = n_1 \,.\, n_2 \,.\, m$ and the ratio of this to the random moment will have the average value

$$M_s/M_r = 1{\cdot}31(n_1 \,.\, n_2)^{1/2}. \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (2)$$

Taking the observed values of $M_{1(\mathrm{sat})}$ and M_{min} as estimates of M_s and M_r respectively we find for a specimen from site 5 that $M_s/M_r = 1{\cdot}5 \times 10^3$ which give $1{\cdot}3 \times 10^6$ domains per cm³. But $M_{1(\mathrm{sat})} \leqslant \frac{1}{2} M_s$ and also $M_{\mathrm{min}} > M_r$, since the directions after the minimum intensity have been reached are not completely random due to the persistence of a very hard component even after treatment in high alternating fields. This will, therefore, be a minimum estimate. A better estimate may be obtained by putting $M_s = 2M_{1(\mathrm{sat})}$ and $M_r = \frac{1}{2} M_{(\mathrm{min})}$ which gives $M_s/M_r = 5{\cdot}9 \times 10^3$ and $20{\cdot}4 \times 10^6$ domains per cm³.

Examination of a thin section of this specimen shows that although there is wide range in the size of visible magnetic grains, about 80% of the magnetic material is contained in grains of size 25–150 μ and over 99% in the range 25–175 μ. A modal analysis of this section shows that the magnetic minerals occupy more than 10% by volume of the rock. Assuming that the material of grain size $> 25\,\mu$ is responsible for the observed minimum intensity and that these grains are spheres of average diameter $100\,\mu$, there will be $1{\cdot}9 \times 10^5$ per cm³. From the estimates of the total number of domains, this gives the order of 10 or 100 domains per particle and suggests that the minimum intensity is due to the presence of multi-domain particles. For the reasons just mentioned the estimate of ten domains per particle will be a minimum. The estimate of the order of 100 is likely to be more correct and there is no reason to regard this as a maximum estimate.

Since this paper was written 30 specimens from these basalts have been thermally ' demagnetized ', by heating to 610°c (more than 50° higher

than the Curie Points of the iron minerals they contain) in nitrogen in a non-magnetic oven followed by cooling in zero field. After this treatment the specimens still retained an appreciable moment. The values, which are to be compared with the minimum intensities obtained by demagnetization in alternating magnetic fields (table 2, column 3) are these: site **1**, 0·11; **2**, 0·64; **3**, 1·67; **4**, 0·22; **5**, 0·23; **6**, 0·16; **7**, 0·43; **8**, 0·01. The values are an average of about four specimens per site and are given in units of 10^{-3} e.m.u./cm^3. As with demagnetization by alternating fields these small residual moments could result from instrumental imperfections due, for instance, to incomplete cancellation of the earth's field. The de-gaussing coils have a central space of about 8000 cm^3 which is uniform to about 1 part in 10^5 and the oven, which has a working space rather less than this, holds about 20 specimens spaced sufficiently far apart to avoid mutual interference from their moments. The directions in each batch after treatment are random so that it appears that the residual field in the oven has negligible aligning force on the specimens. The important point in the present context is that the magnitudes of the moments, M_{min}, are the same whether they are obtained by thermal or alternating field demagnetization; it would be a remarkable coincidence if instrumental imperfections in two different methods produce such closely comparable effects. These results therefore provide strong evidence confirming the view that the minimum intensity is an inherent property of these basalts and results from the statistical nature of the remanent magnetization.

The accuracy of the procedure of removal of the unwanted secondary magnetizations by alternating magnetic fields will be limited by the presence of the minimum intensity, in those cases where the magnitude of the soft component is large, since the minimum intensity of this may exceed the magnitude of the hard component. However, since the minimum intensity directions are random, the average of several specimens will give an estimate of the direction of the primary component although the individual specimen directions will be widely scattered. This is probably the case at site **3** where even after removal of the secondary component the directions are widely scattered but the mean is significant and agrees with the direction of the primary component obtained at the other sites. In rocks containing multi-domain particles which have become demagnetized over long periods of time a minimum intensity will always be present and may be the cause of the random scatter in the directions of NRM observed in specimens taken from the same sample of lava and which in our experience is usually greater than the experimental error. The presence of small temporary components could also explain this.

§ 8. Ferromagnetic Components

In addition to the grain sizes notes above there are a large number of very small grains of the order of a micron in diameter and a small portion of the stable NRM may be due to these. The directions of magnetization

do not become completely random (figs. 2 (*d*) and 3 (*d*)) even after treatment in high alternating magnetic fields suggesting that there is an exceedingly hard component present and these grains, acting as single domains, may be responsible for this. They could not be responsible for all the stable NRM since they represent less than 0·1% (by volume) of the total ferromagnetic content. It seems therefore that both the primary and secondary magnetizations reside in grains which, because of their size and the considerations in the previous sections, are multi-domain. The evidence presented in this paper is consistent with the view that in the natural rock these grains are for the most part either demagnetized and contribute only their minimum intensity, or possess a secondary magnetization, but that they contain within themselves regions of much higher magnetic stability responsible for the primary NRM which, as was shown in § 5, is probably thermo-remanent in origin. It may be noted in fig. 6 that the IRM induced in a steady field of 300 oersteds ($M_{I(300)}$) is removed by alternating magnetic fields of the same strength. Moreover, if $M_{I(300)}$ is induced in a specimen previously possessing a TRM ($M_{U(0\cdot6)}$) which has been treated in alternating fields in excess of 300 oersteds (say 500 oersteds), the underlying TRM remains unaffected after the removal of $M_{I(300)}$. The two components ($M_{I(300)}$, and $M_{U(0\cdot6)}$ after treatment in 500 oersteds $H_p \sim$) are independent of one another and are probably related to two different ferromagnetic components in these basalts, the former perhaps to the multi-domain ferromagnetic grains and the latter to high stability regions within these.

An estimate of the fractional content of grains represented by high stability regions may be obtained from considering the demagnetization of $M_{U(0\cdot6)}$ (fig. 6). The softer component which makes up about 90% of the intensity is largely removed in 500 oersteds, and it is this component which is likely to become demagnetized in the natural rock over a long period of time, or which has acquired a secondary magnetization. In fields above this value a hard component, which is 6% of $M_{U(0\cdot6)}$ in the highly unstable Robertson basalt (site **3**) and 15% of $M_{U(0\cdot6)}$ in the stable basalt at site **8**, still remains. It seems that these high stability regions occupy approximately 10% of the ferromagnetic grains.

Verhoogen (1959) has recently suggested in general terms that such high stability regions within ferromagnetic grains are due to structural dislocations. Another possibility is that intergrowths are responsible; the fine scale intergrowths contributing the hard magnetic component and the coarser-grained matrix providing the softer component. Dislocations are more likely to occur in rapidly cooled rocks whereas slow cooling will favour the development of suitable intergrowths. It is noteworthy that in extensive work on hypabyssal dolerite intrusives (Almond *et al.* 1956, Irving 1956, Graham and Hales 1957, Bull and Irving 1960, Creer *et al.* 1959) and plutonic intrusions of gabbro (Gough and Niekerk 1959, Khan 1960) few cases of instability have been found, whereas instability in basalts is very common (Brynjólfsson 1957

Creer 1958, Khan 1960, this paper). In dolerite and gabbro the ferromagnetic grains are larger than those in basalt and it is clear that there is no simple correlation between stability and the overall size of the iron mineral grains. It seems more likely that the question of their internal subdivision is of more importance. It seems not unlikely that the explanation for the hard and soft magnetic components which are found not only in these basalts discussed in this paper but also in most other igneous rocks studied is to be sought in the exsolved phases of differing grain size in ferromagnetic grains.

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REFERENCES

ALMOND, M., CLEGG, J. A., and JAEGER, J. C., 1956, *Phil. Mag.*, **1**, 771.
AS, J. A., and ZIJDERVELD, J. D. A., 1958, *Geophys. J.*, **1**, 308.
BOESEN, R., IRVING, E., and ROBERTSON, W. A., 1961, *Proc. roy Soc. Qd*, **94** (in the press).
BRYNJÓLFSSON, A., 1957, *Advanc. Phys.*, **6**, 247.
BULL, C., and IRVING, E., 1960, *Geophys. J.*, **3**, 211,
CREER, K. M., 1957, *Phil. Trans. roy. Soc.* A, **250**, 130 ; 1958, *Ann. Geophys.*, **14**, 373 ; 1959, *Geophys. J.*, **2**, 261.
CREER, K. M., IRVING, E., and NAIRN, A. E. M., 1959, *Geophys. J.*, **2**, 306.
DAVID, T. W. E., 1950, *Geology of the Commonwealth of Australia*, **1** (Arnold).
FISHER, R. A., 1953, *Proc. roy. Soc.* A, **217**, 295.
GOUGH, D. I., and NIEKERK, C. B., 1959, *Phil. Mag.*, **4**, 126.
GRAHAM, K. W. T., and HALES, A. L., 1957, *Advanc. Phys.*, **6**, 149.
IRVING, E., 1956, *Pap. roy. Soc. Tas.*, **90**, 157.
IRVING, E., and GREEN, R., 1957, *Mon. Not. R. astr. Soc. geophys. Suppl.*, **7**, 347.
KHAN, M. A., 1960, *Geophys. J.*, **3**, 45.
PARRY, J. H., 1957, Thesis, University of Cambridge.
RAYLEIGH, LORD, 1880, *Phil. Mag.*, **10**, 77 ; 1919, *Ibid.*, **37**, 321.
STOTT, P. M., and STACEY, F. D., 1960, *J. geophys. Res.*, **65**, 2419.
VERHOOGEN, J., 1959, *J. geophys. Res.*, **64**, 2441.
WATSON, G. S., 1956, *Mon. Not. R. astr. Soc. geophys. Suppl.*, **7**, 160.